

Ch-11-

Sets.

Defⁿ:- A Set is a well defined collection of objects.

Note:- i) Sets are usually denoted by Capital Letters. (A, B, C, ...)
 (ii) The elements of sets are represented by small letters (a, b, c, ...).

N- Natural, W- Whole, I/Z- Integer, Q- Rational, Q^c- Irrational
 R- Real No.

→ If (small) "a" is an element of Set "A", then we say

"a belongs to A"
 We write it as " $a \in A$ "

→ If "a" is not an element of Set "A", then we say.

"a does not belong to A" → " $a \notin A$ "

Representation of a Set:-

i) Roster or Tabular form:- In it, all the elements of sets being separated by "," & enclosed with "{ }".

eg:- i) Set of all N < 5 ⇒ {1, 2, 3, 4}

ii) " " all Vowels ⇒ {a, e, i, o, u}

ii) Set Builder form: In it, all the elements of a set possess a single common property which is not possessed by any element outside the set.

eg: a) $\{x | x \text{ is a } N < 5\}$ b) $\{x | x \text{ is vowel in Eng. Alph.}\}$

Q. Write the following Set in Roster form:

i) $A = \{x | x \text{ is an integer } \& -3 < x < 7\}$

Sol: $\{-2, -1, 0, 1, 2, 3, 4, 5, 6\}$

ii) $B = \{x | x \text{ is 2-digit } N, \text{ Sum} = 8\}$

Sol: $\{17, 26, 35, 44, 53, 62, 71, 80\}$

iii) $C = \{x | x \text{ is letters in word 'BETTER'}\}$

Sol: $\{B, E, T, R\}$

Note: The Roster form of a set is unique, but Set builder is not unique.

eg: $\{1, 2\} \rightarrow A = \{x | x \text{ is } N < 3\}$

$B = \{x | x \text{ is the sol}^n \text{ of eqn } (x-1)(x-2) = 0\}$

eg: $\{3, 6, 9, 12\} \rightarrow A = \{x | x \text{ is multiple of 3, } < 15\}$

$B = \{x | x = 3n, n \in N < 5\}$

→ Finite Set

Empty Set:- A Set which does not contain any element.

→ Also known as:- Null Set, Void Set.

→ We represent it by $\emptyset$ or $\{\}$ ← empty brackets.

eg:- $\{x | x \in \mathbb{N} \text{ \& } 1 < x < 2\} \rightarrow \{\}$

Finite & Infinite Set:-

Finite:- A Set which is Empty or contain finite no. of element.

Infinite:- " " " don't contain finite no. of element.

eg:- F :- ω be the Set of days of week.

I :- G be the Set of points on the line.

Q. Which are F or I Set:-

i) $\{x | x \in \mathbb{N} \text{ \& } (x-1)(x-2) = 0\} \Rightarrow \text{Finite}$

ii) $\{x | x \in \mathbb{N} \text{ \& } x^2 = 4\} \Rightarrow \text{Finite}$

iii) $\{x | x \in \mathbb{N} \text{ \& } x \text{ is prime}\} \Rightarrow \text{Infinite}$

iv) $\{x | x \in \mathbb{N} \text{ \& } x \text{ is odd}\} \Rightarrow \text{Infinite}$

Equal Set:- Two Sets A & B are said to be equal, if they have exactly same element. And we write $A = B$.

→ Otherwise, the Sets are said to be unequal, write it as $A \neq B$.

eg:- $A = \{1, 2, 3, 4\}$; $C = \{a, b, c, d\}$ } $A \neq B$; $B \neq D$; $A \neq D$
 $B = \{4, 3, 1, 2\}$; $D = \{2, 1, 4, 3\}$ } but $\Rightarrow C \neq A$; $C \neq B$; $C \neq D$.

Quadratic Eqⁿ - $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Q. Find pair of equal sets:-

i) $A = \{0\} \Rightarrow \{0\}$
 $B = \{x | x > 15 \& x < 5\} \Rightarrow \{\} \text{ or } \emptyset$
 $C = \{x | x - 5 = 0\} \Rightarrow \{5\}$
 $D = \{x | x^2 = 25\} \Rightarrow \{-5, 5\}$
 $E = \{x | x \text{ is +ve root of eq}^n \Rightarrow \{+5\}$
 $x = +5 \text{ or } -3 \quad x^2 - 2x - 15 = 0 \Rightarrow$

ii) A, the set of letters in word 'ALLOY' $\Rightarrow \{A, L, O, Y\}$
 B, " " " " " " " 'LOYAL' $\Rightarrow \{L, O, Y, A\}$ equal.

iii) $A = \{n | n \in \mathbb{Z} \& n^2 \leq 4\} \Rightarrow \{-2, -1, 0, 1, 2\}$
 $B = \{x | x \in \mathbb{R} \& x^2 - 3x + 2 = 0\} \Rightarrow \{2, 1\}$ not equal.

$0 \in A$ but $0 \notin B \Rightarrow A \neq B$.

Subsets:- A Set 'A' is said to be subset of Set 'B', if every element of A is also an element of B.

Symbol
Represented by:- $A \subset B$ / $A \subsetneq B$

eg:- $A = \{1, 2, 3, 4\}$; $B = \{1, 2, 4\} \Rightarrow B \subset A$
 $\rightarrow B \subset A$; if whenever $a \in A$, and $a \in B$

$\rightarrow$ empty set is subset of every set.

$\rightarrow$ In equal set if $A = B \Rightarrow A \subset B$ and $B \subset A$.

$\rightarrow$ Every set is subset of itself.

Q. $\phi, A = \{1, 3\}, B = \{1, 5, 9\}, C = \{1, 3, 5, 7, 9\}$

Insect ϵ & ϕ

a) ϕ c B

b) $A \not\subseteq B$

c) A C C

d) B c C

Subsets of Real No.:-

$$N = \{1, 2, 3, 4, \dots\}$$

$$W = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$$

$$\mathbb{Q} = \{x \mid x = \frac{p}{q} \text{ \& } q \neq 0\}$$

N c w

$$W \subset \mathbb{Z}$$

$\mathbb{Z} \subset \mathbb{Q}$

$$d \subset \mathbb{R}$$

$Q^c \subset TR$

NcWcZcOCTR

Interval as Subsets of real no.:-

a) $(1, 3) = \{x \mid x < 1 < x < 3\}$; c) $[1, 3) = \{x \mid 1 \leq x < 3\}$

b) $[1, 3] = \{x \mid 1 \leq x \leq 3\}$; d) $(1, 3] = \{x \mid 1 < x \leq 3\}$

→ Let a, b belong to Real no - and $a < b$,

→ When, the set of real no. is $\{x | a < x < b\}$, called open interval, Denoted as (a, b) .

$\rightarrow$ " " " " " " " "
close interval, " "[a, b].

→ Semi-Open & Semi-Close $\Rightarrow (a, b], [a, b)$.

Power-Set:- The Set of all possible subsets of a Set 'A' is called, Power-Set.

Denoted by:- P(A).

eg- $A = \{1, 2, 3\} \Rightarrow \underbrace{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}}_{\substack{\text{Sub sets.} \\ \#}}$

$\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}$
Power set

eg. $B = \{1, 4\} \Rightarrow \{\emptyset, \{1\}, \{4\}, \{1, 4\}\}$ Set
Powerset $P(B)$

Cardinality of a Set :- the no. of element in a set A is called Cardinality of that Set A .

Denoted by:- $n(A)$

eg. $A = \{a, b, c, d, e\} \Rightarrow n(A) = 5$

Note:- Let A be a set with $n(A) = m$, then $n(P(A))$ is 2^m . (See example of Powerset).

Universal Set:- Denoted by (usually) $= U$
A set which contain all elements of given subsets, also called Basic-Set.

Q. Write all subset of following sets:-
Powerset.

i) $\{a\} \Rightarrow \{\emptyset, \{a\}\}$

ii) $\emptyset \Rightarrow \{\emptyset\}$

$n(P(\emptyset)) = 1$
 $n(\emptyset) = 0$
 $P(\emptyset) = \{\emptyset\}$

iii) $\{1, 2, 3, 4\} \Rightarrow \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{2, 3\}, \{3, 4\}, \{1, 4\}, \{1, 2, 3\}, \{2, 3, 4\}, \{1, 3, 4\}, \{1, 2, 4\}, \{1, 3\}, \{2, 4\}, \{1, 2, 3, 4\}\}$

iv) $\{a, b\} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Q. Write the following as Interval:-

1) $\{x | x \in \mathbb{R}, -4 \leq x \leq 6\} \Rightarrow [-4, 6]$

2) $\{x | x \in \mathbb{R}, -12 < x < -10\} \Rightarrow (-12, -10)$

3) $\{x | x \in \mathbb{R}, 0 \leq x < 7\} \Rightarrow [0, 7)$

4) $\{x | x \in \mathbb{R}, 3 \leq x \leq 4\} \Rightarrow [3, 4]$

Q. Write following in Set-builder form:-

- 1) $(-3, 0)$ $\Rightarrow \{x \mid -3 < x < 0\}$
 2) $[6, 12]$ $\Rightarrow \{x \mid 6 \leq x \leq 12\}$
 3) $(6, 12]$ $\Rightarrow \{x \mid 6 < x \leq 12\}$
 4) $[-23, 5)$ $\Rightarrow \{x \mid -23 \leq x < 5\}$

Q. Given $\Rightarrow A = \{1, 3, 5\}$ $B = \{2, 4, 6\}$ $C = \{0, 2, 4, 6, 8\}$
 $U = ?$

- a) $\{0, 1, 2, 3, 4, 5, 6\}$ b) $\emptyset$
 ✓ $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ d) $\{1, 2, 3, 4, 5, 6, 7, 8\}$

Operation on Set:-

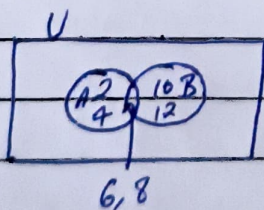
1) Union of Sets:- Let, A & B any 2 sets, the union of A & B is the set which contains all the elements of A & B.

$\rightarrow$ The common elements been taken only once.

$\rightarrow$ Symbol:- "U" & written as $A \cup B$.

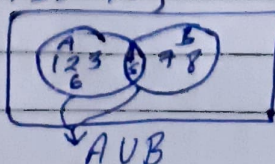
Q. if $A = \{2, 4, 6, 8\}$ & $B = \{6, 8, 10, 12\}$; find $A \cup B$ and make Venn Diagram.

$A \cup B$ = $\{2, 4, 6, 8, 10, 12\}$



eg:- $A = \{1, 2, 3, 4, 5, 6\}$; $B = \{4, 5, 7, 8\}$

$A \cup B$ = $\{1, 2, 3, 4, 5, 6, 7, 8\}$



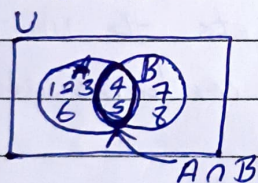
→ Properties of Union:-

- i) Commutative $\Rightarrow A \cup B = B \cup A$
- ii) Associative $\Rightarrow (A \cup B) \cup C = A \cup (B \cup C)$
- iii) $A \cup \phi \Rightarrow A$
- iv) $A \cup U \Rightarrow U$
- v) $A \cup A \Rightarrow A$

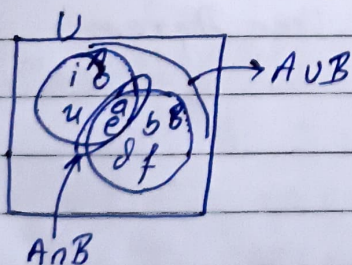
2) Intersection of Sets:- The intersection of set A & B is the set of all element which are common in both A & B.

→ Symbol = " $\cap$ ", written $\Rightarrow A \cap B$.

eg:- $A = \{1, 2, 3, 4, 5, 6\}$; $B = \{4, 5, 7, 8\}$
 $A \cap B = \{4, 5\}$



d. $A = \{a, e, i, o, u\}$ & $B = \{b, a, e, d, f\}$
 find $A \cup B = \{a, e, i, o, u, b, d, f\}$
 $A \cap B = \{a, e\}$



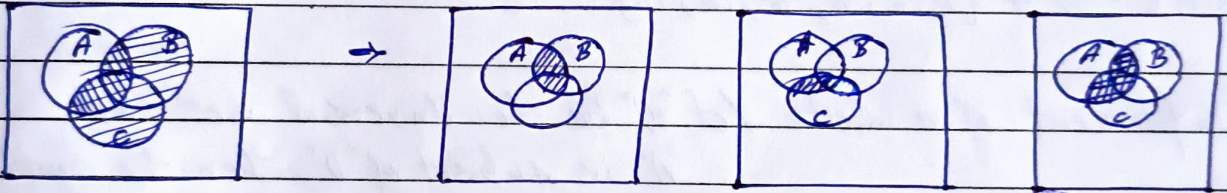
→ Properties of Intersection:-

- i) Commutative $\Rightarrow A \cap B = B \cap A$
- ii) Associative $\Rightarrow (A \cap B) \cap C = A \cap (B \cap C)$
- iii) $A \cap \phi = \phi$
- iv) $A \cap U = A$
- v) $A \cap A = A$

3) Distributive Law:-

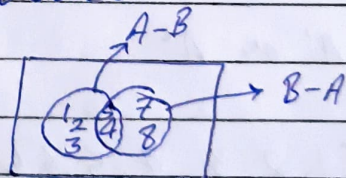
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$



* **Difference of Set**:- The diff. of the sets A and B in this order is the set of order.

$$A - B = \{x / x \in A \text{ \& } x \notin B\}$$



$$A = \{1, 2, 3, 4, 5, 6\} \quad ; \quad B = \{4, 5, 6, 7, 8\}$$

$$A - B = \{1, 2, 3\} \quad ; \quad B - A = \{7, 8\}$$

d. if $A = \{3, 6, 9, 12, 15, 18, 21\}$

$$B = \{4, 8, 12, 16, 20\}$$

$$C = \{4, 6, 8, 10, 12, 14, 16\}$$

$$D = \{5, 10, 15, 20\}$$

find:- i) $A \cup B \Rightarrow \{3, 6, 9, 12, 15, 18, 21, 4, 8, 16, 20\}$

ii) $A \cap C \Rightarrow \{6, 12\}$

iii) $C \cap D \Rightarrow \{10\}$

iv) $A \cap D \Rightarrow \{15\}$

v) $A - B \Rightarrow \{3, 6, 9, 15, 18, 21\}$

vi) $B - A \Rightarrow \{4, 8, 16, 20\}$

vii) $B - C \Rightarrow \{20\}$

viii) $B - D \Rightarrow \{4, 8, 12, 16\}$

ix) $D - B \Rightarrow \{5, 10, 15\}$

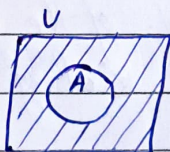
x) $D - C \Rightarrow \{5, 15, 20\}$

xi) $A - C \Rightarrow \{3, 9, 15, 18, 21\}$

xii) $A - D \Rightarrow \{3, 6, 9, 12, 18, 21\}$

xiii) $C - D \Rightarrow \{4, 6, 8, 12, 14, 16\}$

5) Complement of a Set:- Let " U " be the Universal set and " A " is subset of " U ", then the complement of " A " is the set of all element of " U " which are not the element of " A ".



Write A' or A^c

eg:- $U = \{1, 2, 3, 4, 5, 6\}$

$A = \{5, 6\}$

$A^c = \{1, 2, 3, 4\}$

Q.1- $U = \{1, 2, 3, 4, 5, 6\}$

$A = \{2, 3\}$; $B = \{3, 4, 5\}$

find A' & B'

$(A \cup B)' = A' \cap B'$

$(A \cup B)' = \{1, 6\}$

$A \cup B = \{2, 3, 4, 5\}$

$A' = \{1, 4, 5, 6\}$

$(A \cap B)' = \{1, 6\}$

$A' \cap B' = \{1, 6\}$

$B' = \{1, 2, 6\}$

Complement Properties:-

i) Complement Law:-

$$a) A \cup A' = U$$

$$b) A \cap A' = \phi$$

ii) Law of Double Complements:-

$$(A')' = A$$

iii) Law of empty set & Universal set:-

$$\phi = U$$

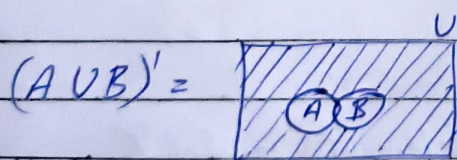
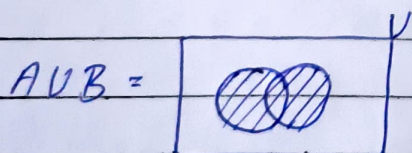
$$U' = \phi$$

iv) De Morgan's Law:-

$$a) A \cup B' = A' \cap B'$$

LHS:- $(A \cup B)'$:-

RHS:-



Q. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$; $A = \{1, 2, 3, 4\}$
 $C = \{3, 4, 5, 6\}$; $B = \{2, 4, 6, 8\}$

find:-
 i) $A' = \{5, 6, 7, 8, 9\}$
 ii) $B' = \{1, 3, 5, 7, 9\}$
 iii) $(A \cup C)' = \{7, 8, 9\}$
 iv) $(A \cup B)' = \{5, 7, 9\}$
 v) $(A')' = \{1, 2, 3, 4\}$
 vi) $(B - C)' = \{1, 4, 6, 7\}$
 $\Rightarrow \{2, 8\} \Rightarrow \{1, 3, 4, 5, 6, 7, 9\}$

Q. a) $A \cup A' = U$ c) $A \cap A' = \emptyset$
 b) $\emptyset \cap A = A'$ d) $U' \cap A = \emptyset$

6) Cartesian Product of sets:- Given 2 non empty Set P & Q , the Cartesian Product $P \times Q$ is the Set of all $\boxed{P \times Q = \{(p, q) \mid p \in P \text{ \& } q \in Q\}}$ order pair of element.

$P = \{a, b\}$, $Q = \{1, 2, 3\}$

$P \times Q = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$

$Q \times P = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$

Note:- 1) $(x, y) = (3, 4) \Rightarrow x = 3 \text{ \& } y = 4$
 2) Don't follow commutative Law.

3) if there are 'n' element in Set 'A' & 'm' elements in Set 'B',
 then Cardinality of $A \times B \Rightarrow \overset{\text{mat.}}{n \times m} \Rightarrow n(A) = n \rightarrow n(A \times B) = n \times m$
 $n(B) = m$

4) if A & B are non-empty sets and either A or B is an ∞ set then $\boxed{A \times B = \{\infty\}}$.

5) if one of A or B is an empty set ($\emptyset$), then $\boxed{A \times B = \emptyset}$.

Q. if $(x+1, y-2) = (3, 1)$; find x, y :-

$$x+1=3$$

$$\boxed{x=2}$$

$$y-2=1$$

$$\boxed{y=3}$$

Q. if $P = \{1, 2\}$; $P \times P = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

$$P \times P \times P = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)\}$$

Q. if $A = \{c, d\}$; $B = \{1, 2\}$

$$A \times B \times A = \{(c, 1, c), (c, 1, d), (c, 2, c), (c, 2, d), (d, 1, c), (d, 1, d), (d, 2, c), (d, 2, d)\}$$

Q. if $A = \{1, 2, 3\}$; $B = \{3, 4\}$; $C = \{4, 5, 6\}$

$$B \cup C = \{3, 4, 5, 6\}$$

$$i) A \times (B \cap C) = \{1, 4\}, \{2, 4\}, \{3, 4\}$$

$$iii) A \times (B \cup C) = \{(1, 3), (1, 4), (1, 5), (1, 6), (2, 3), (2, 4), (2, 5), (2, 6), (3, 3), (3, 4), (3, 5), (3, 6)\}$$

$$ii) (A \times B) \cap (A \times C) = \{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4) \cap (1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6)\} \\ \Rightarrow \{(1, 4), (2, 4), (3, 4)\}$$

$$iv) (A \times B) \cup (A \times C) = \{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4), (1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6)\}$$

→ Relation:- Relation "R" from non-empty A to non-empty set "B" is a subset of cartesian product A × B.

→ The subset is derived by describing a relationship b/w the first & second element of ordered pair A × B.

→ The second element is called image of first element.

→ Domain:- the set of ^{all} first element of the ordered pair in a relation are from a set "A" to a set "B" is called the domain of Relation "R".

→ Co-Domain & Range:- The set of all second element in a relation are from set "A" to set "B" is called range of Relation "R".

→ The whole set "B" is called Co-domain of Relation "R".

Note:- Range $\subset$ Co-domain.

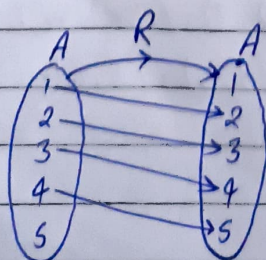
⇒ Representation of Relation

$$A = \{1, 2, 3, 4\} ; R: A \rightarrow A$$

i) Roster form:- $R = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$

ii) Set Builder form:- $\{(x, y) \mid y = x + 1 \ \& \ x \in A, \text{ where } x < 5\}$
or where $x = 1, 2, 3, 4$
or $x \in \mathbb{N} ; x < 5$

iii) Line Diagram:-

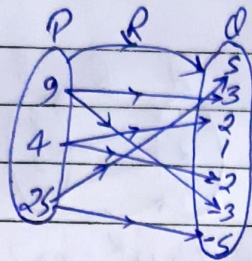


Domain = $\{1, 2, 3, 4\}$

Co-Domain = $A = \{1, 2, 3, 4, 5\}$

Range = $\{2, 3, 4, 5\}$

d. Write roster, Set-builder, domain, Co-domain, range.



Roster :- $R = P \rightarrow Q \Rightarrow \{(9, 3), (9, -3), (4, 2), (4, -2), (25, 5), (25, -5)\}$

Set-Builder :- $\{(x, y) \mid x = y^2 \text{ \& } y \in Q, \text{ where } y \neq 1\}$
 $y = \sqrt{x} \text{ \& } x \in P$

Domain :- $\{9, 4, 25\}$

Co-Domain :- $\{-5, -3, -2, 1, 2, 3, 5\}$

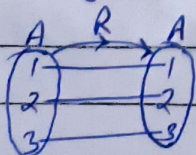
Range :- $\{5, 3, 2, -2, -3, -5\}$

$\Rightarrow$ Types of Relation :-

1) Empty Relation :- Let R be a relation from Set A to a Set B . Then, R is called empty relation if no element of Set A is related to any element of Set B .

2) Universal Relation :- A relation from a Set A to a Set B is called Universal relation if every element of set A is related every element of set B .

3) Identity Relation :- Let R be a relation on Set A , R is called Identity Relation if every element of A is related to itself.

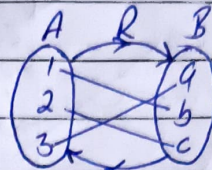


eg :- $R: A \rightarrow A$

$R = \{(1, 1), (2, 2), (3, 3)\}$

4) Inverse Relation: Let R be a relation from a set A to a set B then inverse relation is denoted by R^{-1} and defined as

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$



$$A = \{1, 2, 3\}; B = \{a, b, c\} \Rightarrow R = \{(1, b), (2, c), (3, a)\}$$

$$\Rightarrow R^{-1} = \{(b, 1), (c, 2), (a, 3)\}$$

Q. Let $A = \{1, 2, 3, \dots, 14\}$; Define rel. R from A to A ,

by $R = \{(x, y) \mid 3x - y = 0 \text{ where } x, y \in A\}$,
write Domain, Co-domain & range.

Solⁿ:- $3x - y = 0 \Rightarrow x = 1 \quad 2 \quad 3 \quad 4 \quad \checkmark$ $D = \{1, 2, 3, 4\}$
 $\boxed{3x = y} \quad y = 3 \quad 6 \quad 9 \quad 12 \quad \checkmark$ $C.D = \{1, 2, \dots, 14\}$
 $R = \{(1, 3), (2, 6), (3, 9), (4, 12)\}$ $R = \{3, 6, 9, 12\}$

Q. Define Relation R on the set of Natural No. by.

$$R = \{(x, y) \mid y = x + 5, x \text{ is } N < 4\}$$

$$D = \{1, 2, 3\}$$

$$C.D = \{1, 2, 3, \dots, \infty\} \text{ \{set of N. no.\}}$$

$$y \begin{matrix} 6 & 7 & 8 \end{matrix} \rightarrow R = \{(1, 6), (2, 7), (3, 8)\}$$

$$x \begin{matrix} 1 & 2 & 3 \end{matrix}$$

$$R = \{6, 7, 8\}$$

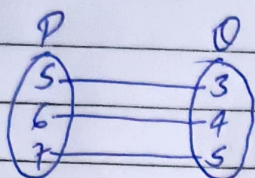
Q. $A = \{1, 3, 2, 5\}$; $B = \{4, 6, 9\}$; Define R from A to B by
 $R = \{(x, y) \mid \text{Diffe. of } x \text{ \& } y \text{ is odd}\}$

$$R = \{(1, 4), (1, 6), (2, 9), (3, 4), (3, 6), (5, 4), (5, 6)\}$$

$$D = \{1, 2, 3, 5\}; R = \{4, 6, 9\}$$

$$CD = \{4, 6, 9\}$$

Q. Write given relation into - i) roster form
ii) Set builder form.



Roster:- $R = \{(5, 3), (6, 4), (7, 5)\}$
S. Builder:- $R = \{(x, y) \mid y = x - 2; x \in N\}$

→ Relation b/w cardinality of A, B, $A \cup B$ & $A \cap B$:-

$n(A)$; $n(B)$; $n(A \cup B)$; $n(A \cap B)$.

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \quad \rightarrow \text{OR} \rightarrow \cup \\ n(A \cap B) &= n(A) + n(B) - n(A \cup B) \quad \rightarrow \text{AND} \rightarrow \cap \end{aligned}$$

Q. In a school there are 20 teachers who teach maths or physics of these 12 teach math and 4 teach math and physics both. How many teach physics.

Solⁿ:- $n(P \cup M) = 20$; $n(M) = 12$; $n(P \cap M) = 4$
 $n(P) = ?$

$$n(P \cup M) = n(M) + n(P) - n(P \cap M).$$

$$20 = 12 + n(P) - 4$$

$$n(P) = 12$$

Q. In a class of 35 students, 24 like cricket, 16 like football. Also each student like atleast one of two game. find how many like both games.

Solⁿ:- $n(C \cup F) = 35$; $n(C) = 24$; $n(F) = 16$; $n(C \cap F) = ?$
 $35 = 24 + 16 - x \Rightarrow x = 40 - 35$

$$\boxed{x = 5}$$

Q. In a survey of 400 students in a school, 100 listed for Apple juice, 150 for Orange juice, 75 for Both juices. find how many taking neither of them.

Solⁿ:- $n(A) = 100$; $n(O) = 150$; $n(A \cap O) = 75$; $n(A \cup O) = ?$

$$n(A \cup O) = 100 + 150 - 75$$

$$n(A \cup O) = 175$$

$$\text{taking neither} \Rightarrow (A \cup O)' \Rightarrow 400 - 175$$

$$n(A' \cap O') \Rightarrow \boxed{225}$$

Q. In a grp. of 65 people 40 like cricket, 10 like Cricket & Tennis, it is given that every person like atleast one game. find no of people who like tennis & also who like only tennis.

Solⁿ:- $n(C) = 40$; $n(C \cap T) = 10$; $n(T) = ?$; $n(C \cup T) = 65$

$$65 = 40 + x - 10$$

$$\boxed{x = 35} \leftarrow n(T)$$

Like Tennis

$$35 - 10 \Rightarrow \boxed{25}$$

Like Tennis only

Q. There are 200 individual with skin disorder, 120 has been exposed to chemical C1, 50 " " " " " C2, 30 " " " to both C1 & C2, find no. of individual exposed to

1) C1 but not C2

2) C2 but not C1

3) C1 or C2 ($C1 \cup C2$)

Solⁿ:- $n(C) = 120$; $n(C_2) = 50$; $n(C \cap C_2) = 30$

$$3) (C \cup C_2) = 120 + 50 - 30$$

$$= \boxed{140}$$

$$1) n(C) - n(C \cap C_2) = 120 - 30 \Rightarrow 90$$

$$2) n(C_2) - n(C \cap C_2) = 50 - 30 \Rightarrow 20$$